

YOUSSEF NASSER

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PROFESSIONAL SUMMARY

Fourth Year Computer Engineering student (GPA 3.4) at AASTMT with hands-on experience across embedded systems, robotics, machine learning, and full-stack software development. Demonstrated ability to architect and deliver end-to-end engineering solutions — from low-level microcontroller firmware to AI-powered vision systems. Actively seeking internship opportunities in Embedded Systems, Robotics, AI/ML, or Software Engineering to apply and expand technical expertise in a professional setting.

EDUCATION

B.Sc. Computer Engineering | Arab Academy for Science, Technology & Maritime Transport (AASTMT) *Oct 2022 – Jan 2027*

GPA: 3.4 / 4.0

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Microprocessors & Embedded Systems, Advanced Computer Networks, Systems Programming, Database Management Systems, Digital Logic Design (VHDL)

American High School Diploma | AASTMT *Sept 2016 – June 2022*

GPA: 4.0 / 4.0

TECHNICAL PROJECTS

RoboSoccerDog — Autonomous Robotic Soccer Player

Tech Stack: Arduino (C), MG996R Servo Motors, ESP32-CAM, Python, OpenCV, TensorFlow, Embedded Systems, Computer Vision

- Designed and built a fully autonomous robotic dog from scratch, integrating Arduino-controlled servo motion with real-time computer vision for soccer ball detection and tracking.
- Implemented a Python-based ML pipeline using TensorFlow and OpenCV on an ESP32-CAM module, enabling the robot to autonomously navigate, aim, and shoot a ball into a goal with no human input.
- Engineered wireless camera-to-microcontroller communication, synchronizing embedded motion control with live visual inference to achieve intelligent closed-loop robotic behavior.

LED Matrix IoT Control System

Tech Stack: Arduino, C++, ESP8266/Wi-Fi, MAX7219 Driver, HTML, CSS, JavaScript, HTTP REST

- Architected a full-stack IoT system enabling real-time remote control of an 8x8 LED matrix display via a custom web dashboard, using the MAX7219 display driver and Arduino C++.
- Implemented HTTP-based bidirectional communication between the embedded device and web frontend, supporting dynamic pattern rendering, animations, and live state updates.
- Demonstrated end-to-end embedded-to-web integration across hardware interfacing, firmware, and responsive UI layers.

E-Waste Image Classification System — Deep Learning

Tech Stack: Python, TensorFlow/Keras, MobileNetV2, Transfer Learning, OpenCV, Scikit-learn, Data Augmentation

- Built a deep learning image classification pipeline to categorize electronic waste across multiple classes using MobileNetV2 with transfer learning, trained on a dataset of 3,000 images with custom data augmentation.
- Achieved 95% classification accuracy on the validation set by fine-tuning pre-trained CNN weights and applying systematic augmentation (rotation, flipping, zoom) to combat overfitting.
- Applied OpenCV for image preprocessing and Scikit-learn for quantitative evaluation (precision, recall, F1-score), producing a fully documented, production-ready model.

Multithreaded Web Crawler

Tech Stack: Python, Scrapy, BeautifulSoup, Tkinter, Multithreading, robots.txt compliance

- Developed a concurrent web crawler with a Tkinter GUI, extracting URLs, titles, metadata, and 500-character content previews from up to 10 links per page.
- Engineered multithreaded crawling architecture delivering 85% faster data retrieval versus single-threaded baseline, while enforcing robots.txt compliance for ethical scraping.

Blood Bank Management System

Tech Stack: C#, Windows Forms, MySQL, CRUD, Role-Based Access Control

- Built a role-based desktop application (admin, staff, donor) for donor registration, blood inventory tracking, and transfusion request management using C# and MySQL.
- Optimized SQL queries and database schema to achieve 90% improvement in real-time blood stock retrieval speed compared to initial implementation.

Gym Management System

Tech Stack: Java, JavaFX, MySQL, CRUD, MVC Architecture

- Engineered a multi-user gym management platform with member registration, class scheduling, and payment tracking, achieving 95% data retrieval efficiency via optimized CRUD operations.
- Applied MVC design pattern with JavaFX for a responsive, event-driven desktop UI backed by a normalized MySQL database.

SKILLS

Programming Languages: C, C++, Python, Java, C#, JavaScript, SQL, VHDL, HTML/CSS

Embedded & Hardware: Arduino, ESP32-CAM, VHDL/ModelSim, Microprocessors, MAX7219, Servo Control, Multisim

AI / ML / Computer Vision: TensorFlow, Keras, MobileNetV2, Transfer Learning, OpenCV, Scikit-learn, Data Augmentation

Web & Backend: React.js, Node.js, Express.js, REST APIs, MySQL, MongoDB

Tools & Platforms: Git, GitHub, Docker, Linux (Ubuntu/Kali/CentOS), AWS, Cisco Packet Tracer, Wireshark, Android Studio, Unity

IDEs & Dev Tools: VS Code, Arduino IDE, IntelliJ, Visual Studio

Soft Skills: Collaboration, Good Communication, Adaptability, time management, problem solving

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- CS50: Introduction to Computer Science — Harvard University (edX) | Algorithms, data structures, C, Python, SQL, web development
- Node.js — CodeWithMosh | Express.js, RESTful APIs, MongoDB, asynchronous programming
- React.js — CodeWithMosh | Hooks, state management, component architecture, API integration
- Java Masterclass — Tim Buchalka (Udemy) | OOP, multithreading, software architecture
- Amazon Web Services (AWS) Workshops — AASTMT | Cloud fundamentals, deployment
- Linux (CentOS, Kali Linux, Ubuntu) — ITI | System administration, CLI, networking